STAGE 1 - SKYWATCH

CipherNex-301 CTF | Domain: OSINT / Reconnaissance | Difficulty: Easy

Stage ID	STG-01	**Points**	100
Domain	OSINT / Reconnaissance	**Flag Format**	CIPHERNEX{snake_case_string}

SCENARIO

A decommissioned experimental satellite, CipherNex-301, has unexpectedly reactivated and begun broadcasting fragmented telemetry toward a private ground station. Intelligence suggests it may be functioning as a covert relay. Northstar Orbital Systems has released a public mission status bulletin along with archived ground station logs covering the spacecraft's operational history. As the analyst assigned to this recovery effort, your job is to identify the satellite, trace its final transmissions, and recover the data hidden within them before the next orbital pass.

OBJECTIVE

Analyze the provided documents and ground station logs to identify the spacecraft, understand its communication parameters, and recover the hidden flag from its final transmission data.

PROVIDED FILES

- **CNX-OPS-301-ST01_MissionBulletin.pdf** - public mission status bulletin
- **ground_station_full_mission.log** - consolidated ground station log (plain text)

SUGGESTED TOOLS

- Web browser
- Public satellite tracking registries (e.g. CelesTrak, N2YO)
- pdfinfo / exiftool (PDF metadata inspection)
- Any log viewer
- A hex-to-ASCII decoder (CyberChef, Python, or similar)

YOUR TASK

1. Review the mission bulletin and identify what information about the satellite is public versus redacted.
2. Use public satellite tracking resources to cross-reference and confirm the spacecraft's identity.

3. Inspect the mission bulletin for any information not visible in the document body itself.
4. Examine the ground station logs, paying close attention to the spacecraft's final active transmission window.
5. Identify and reconstruct the data embedded across that final sequence of transmissions.
6. Decode the reconstructed data to obtain the flag.

FLAG SUBMISSION

Flags follow the exact-match format shown below. Submit your recovered flag through the CTFd platform on the Stage 1 - Skywatch challenge page.

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CIPHERNEX{snake_case_string}
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HINTS (OPTIONAL)

- *Hint 1:* Look at the sky, not the ground.
- *Hint 2:* Amateur radio operators archive downlinks and not everything in a document is visible when you first open it.
- *Hint 3:* The final transmissions weren't random. Order matters.

PROGRESSION

You must successfully capture and submit the Stage 1 flag before Stage 2 (Broken Beacon) is unlocked on the CTFd platform. Stage 2 will build directly on communication parameters you uncover here, so keep track of everything you find along the way.